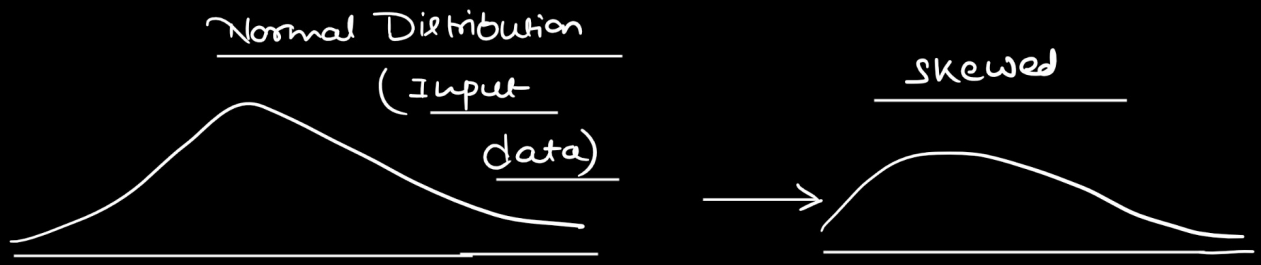


Challenge in Production ML System



① Data Drift

Input Data
distribution changes
over the time

①

market trend
shift

②

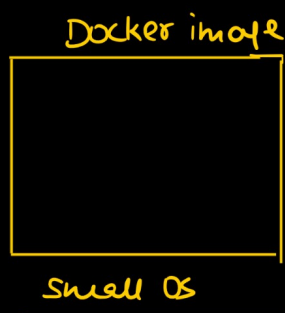
user behavior
changes

② Model Drift → Model performance decreases
with time

Patterns evolve with time

③ Deployment Complexity → API, Docker Containers,

CI/CD Pipeline,
cloud infrastructure



→ code will be
running in an AWS,
appropriate way Azure,
GCP

Notebook → accuracy

4

Monitoring

accuracy

numeric data

↳ KS test

categorical

data

② latency

③ Drift → chi-square test

④ Summary statistics

⑤ Model-based

drift detection

DA → PowerBI, Excel

ETL → Extract

Data collection

(csv, excel)

Transform

Data cleaning/EDA

(Python

coding)

Load

→ for the next stage